SKILLORA

AI-powered Resume Builder and Job Recommendation Bot

Abstract

Skillora is an AI-powered Resume Builder and Job Recommendation System designed to help users discover suitable job opportunities through intelligent skill analysis and automated matching. The system delivers an end-to-end workflow in which users upload resumes through a Telegram bot interface, enabling smooth interaction and high accessibility. Uploaded resumes are parsed to extract structured textual information, which is then analyzed using a keyword-driven and organized skill extraction approach to identify real-world employable competencies rather than isolated technologies alone.

The extracted skills are matched against job requirements using a similarity-based ranking mechanism, ensuring that the job recommendations remain relevant, explainable, and aligned with the user profile. Skillora implements a hybrid job data acquisition strategy, combining mock job datasets for reliability and fallback support with real-world job listings fetched from external APIs. To boost performance and reduce repeated API dependency, the system caches job data in a relational database using SQLAlchemy ORM, enabling fast retrieval and efficient reuse of recently fetched job listings.

The overall architecture follows clean design principles and modular separation between bot handling, resume processing, skill extraction, job matching, database management, and notification services. This modular design strengthens scalability, maintainability, and adaptability for future upgrades such as email notifications, advanced machine learning models, and analytics-driven insights. By integrating automation, explainable matching logic, and real-world data sources, Skillora demonstrates a practical application of backend engineering and AI techniques to solve employment-oriented challenges effectively.